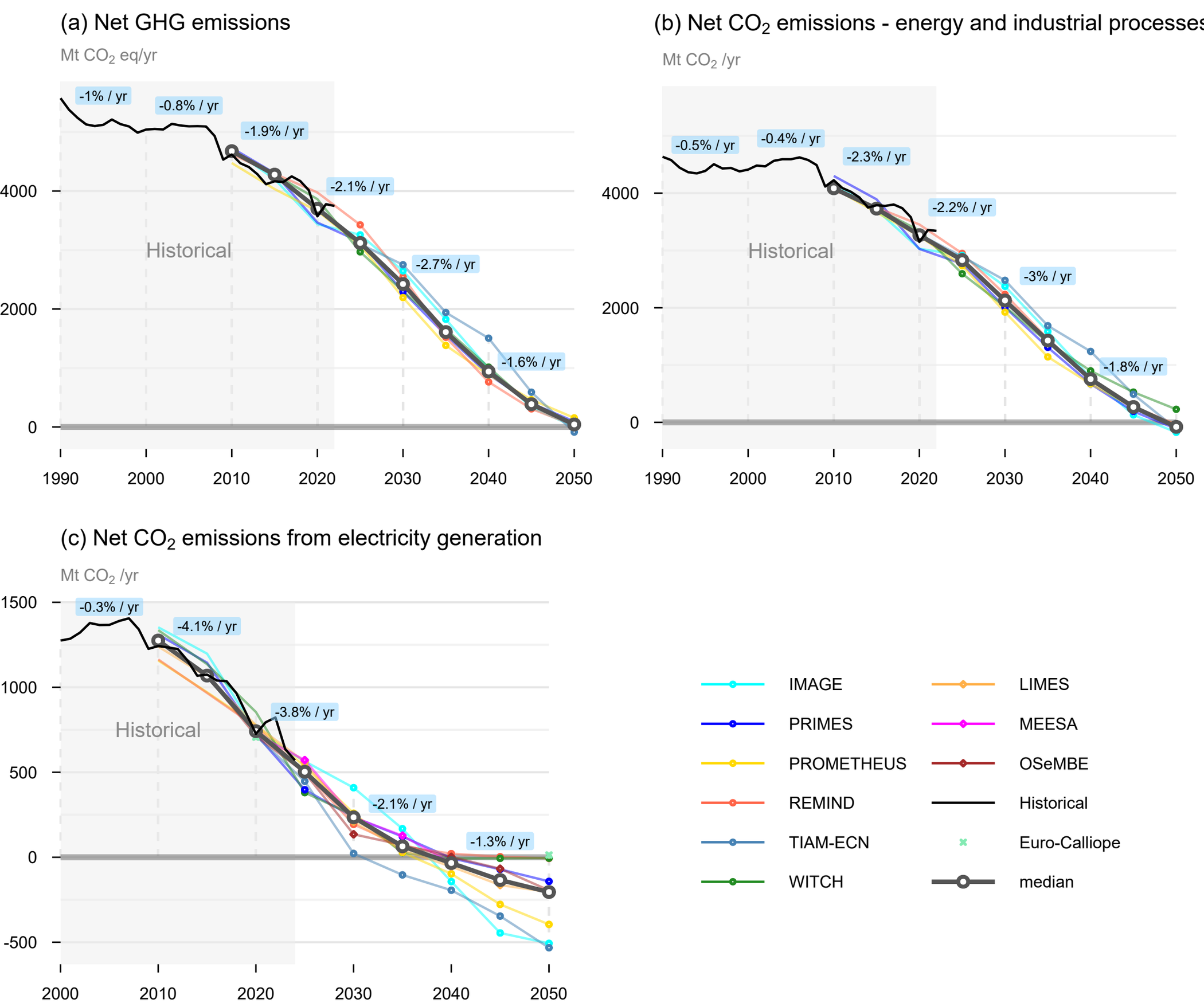
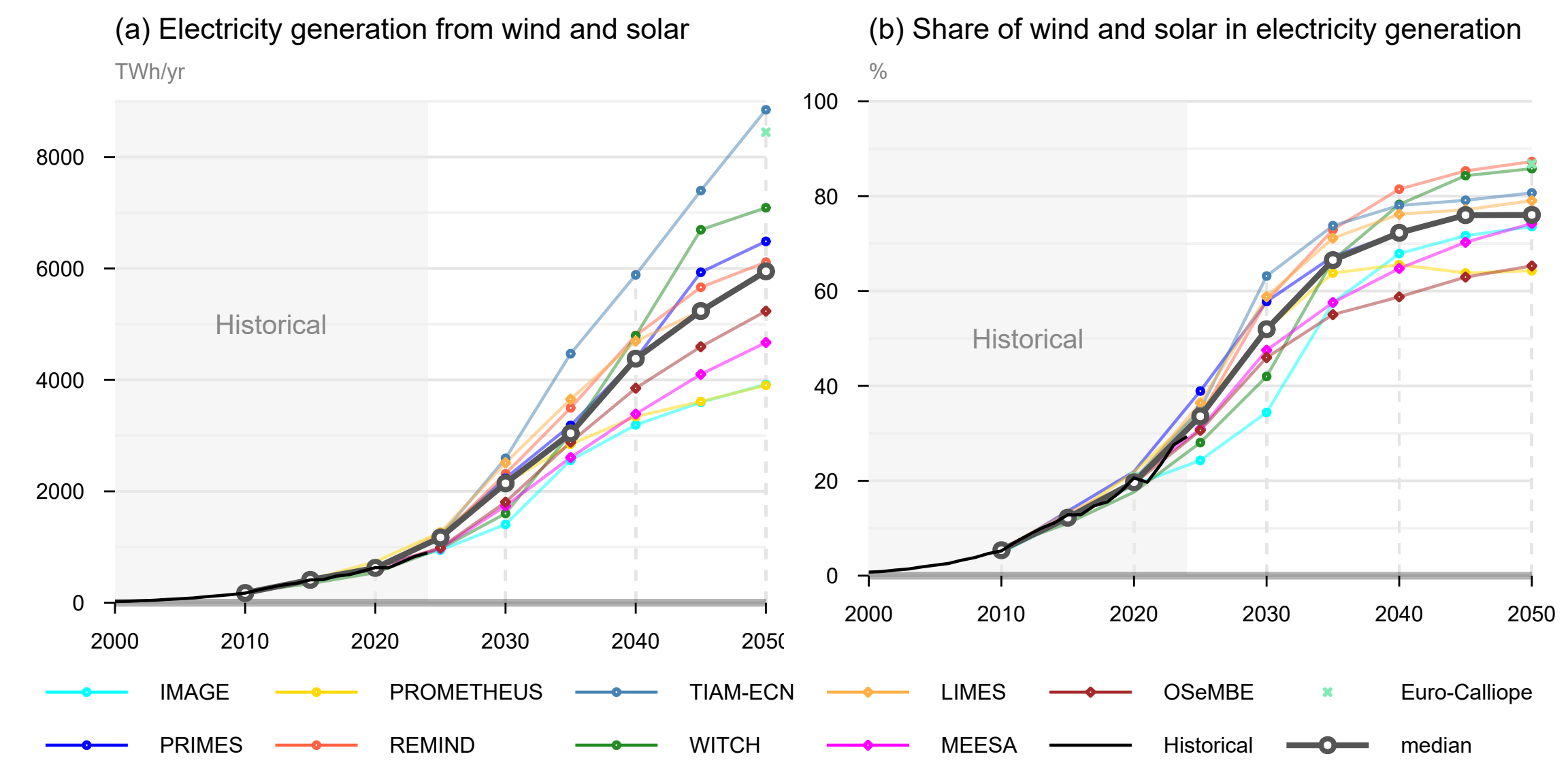
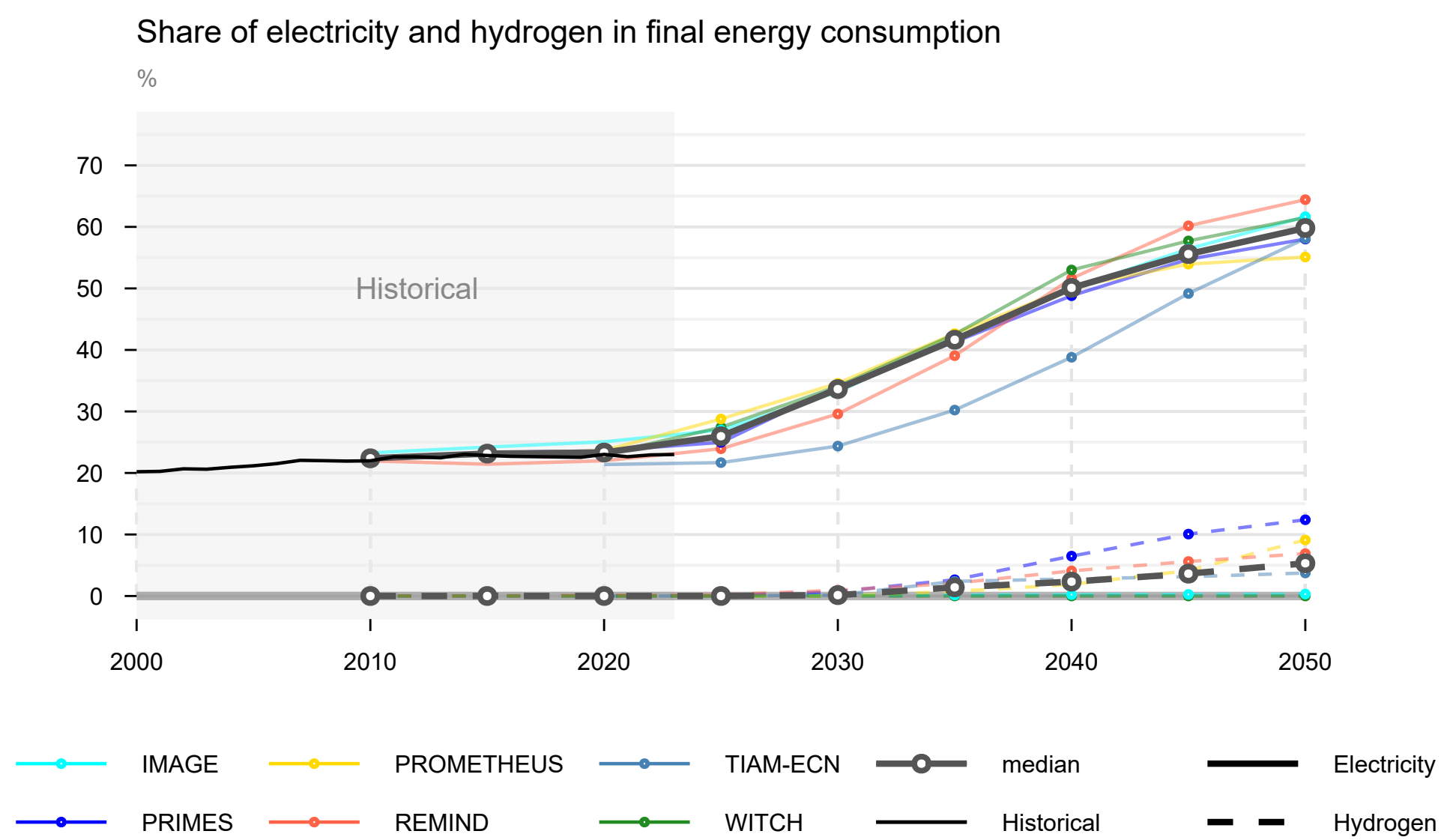
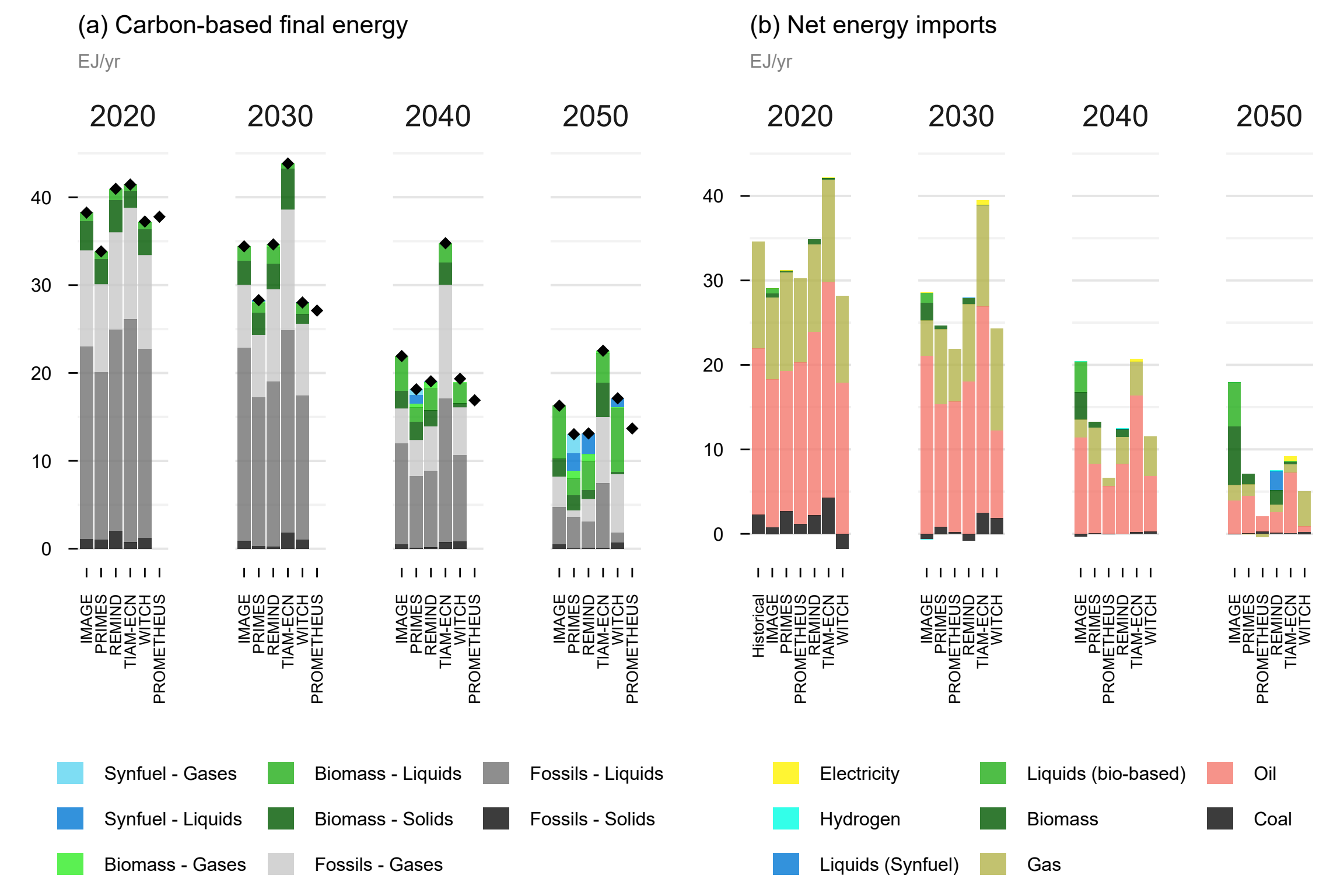
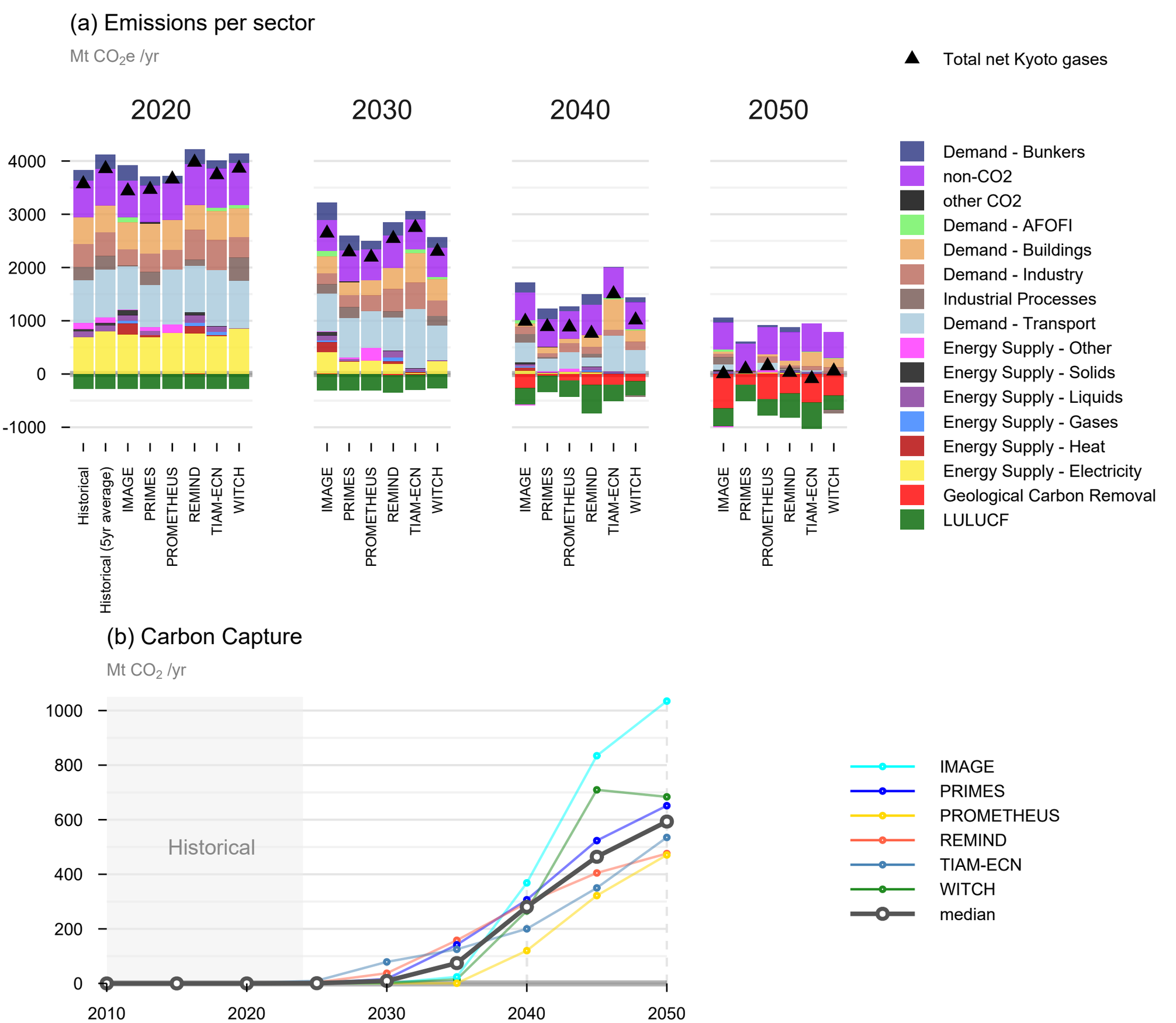
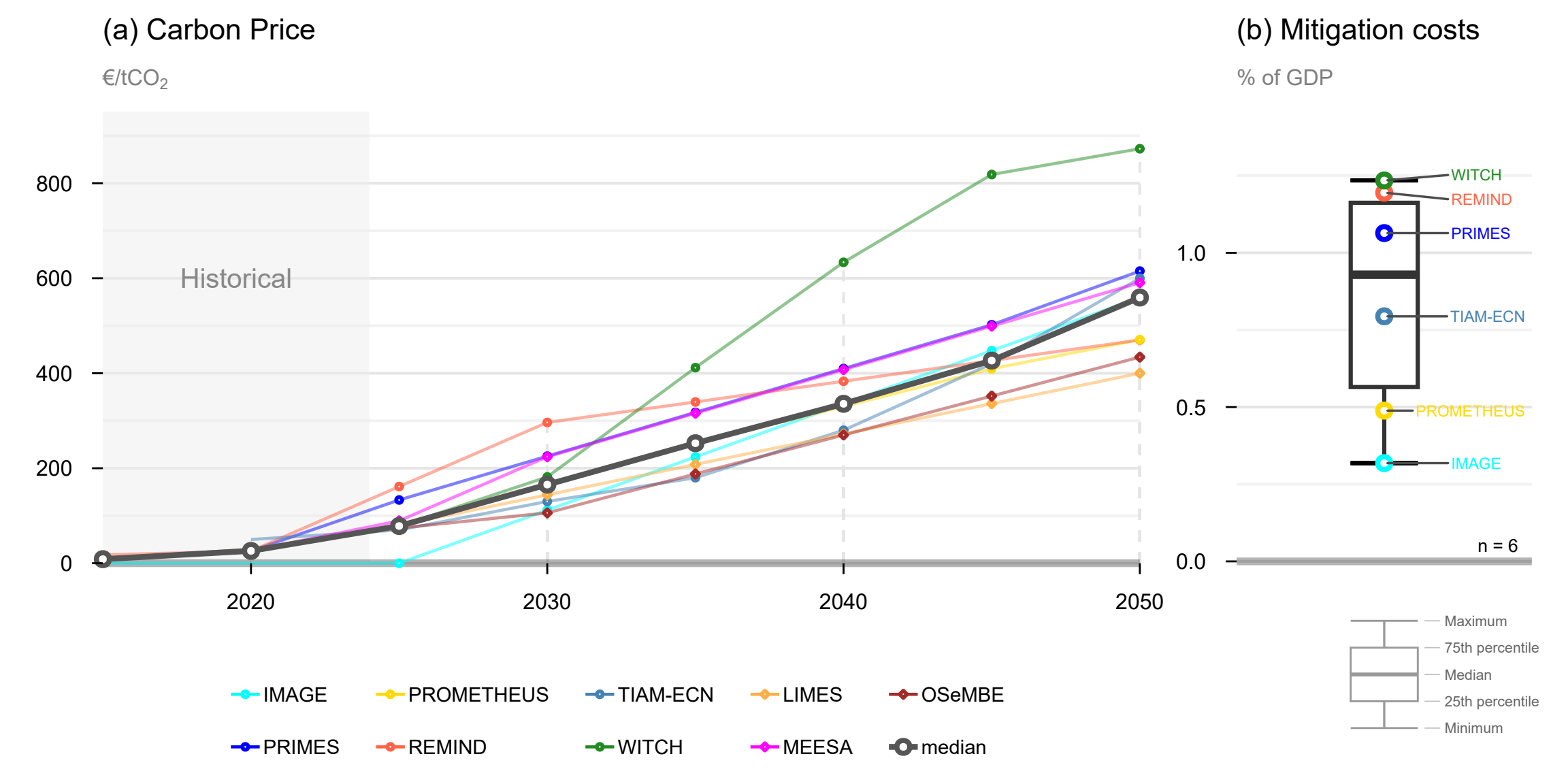

(a) Net GHG emissions
Mt CO2 eq/yr
-1% / yr
-0.8% / yr
-1.9% / yr
-2.1% / yr
-2.7% / yr
-1.6% / yr
Historical
4000
2000
0
1990
2000
2010
2020
2030
2040
2050
(b) Net CO2 emissions - energy and industrial processes
Mt CO2 /yr
-0.5% / yr
-0.4% / yr
-2.3% / yr
-2.2% / yr
-3% / yr
-1.8% / yr
Historical
4000
2000
0
1990
2000
2010
2020
2030
2040
2050
(c) Net CO2 emissions from electricity generation
Mt CO2 /yr
-0.3% / yr
-4.1% / yr
-3.8% / yr
-2.1% / yr
-1.3% / yr
Historical
1500
1000
500
0
-500
2000
2010
2020
2030
2040
2050
IMAGE
PRIMES
PROMETHEUS
REMIND
TIAM-ECN
WITCH
LIMES
MEESA
OSeMBE
Historical
Euro-Calliope
median

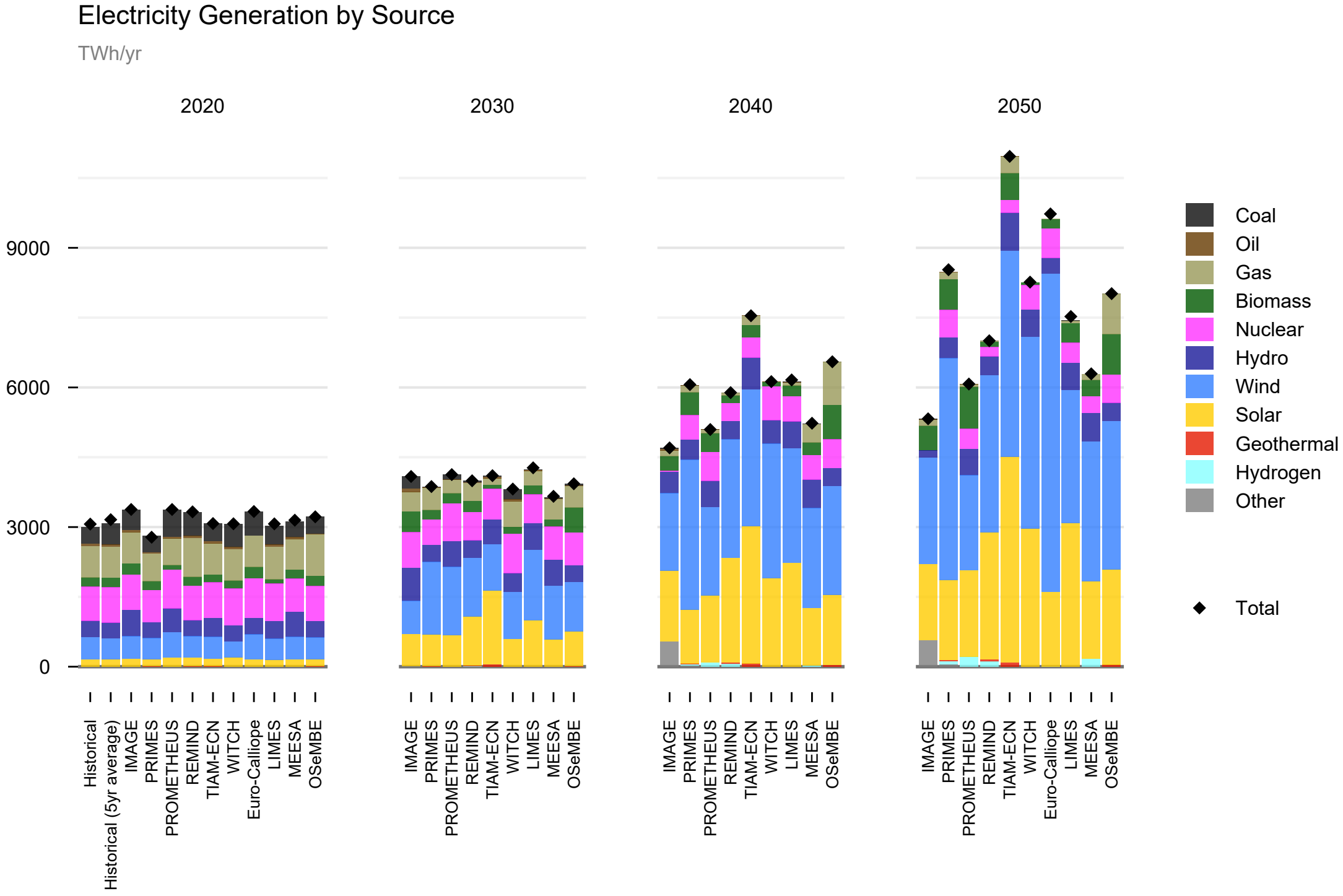

(a) Electricity generation from wind and solar
TWh/yr
8000
6000
4000
2000
0
Historical
2000
2010
2020
2030
2040
2050
(b) Share of wind and solar in electricity generation
%
100
80
60
40
20
0
Historical
2000
2010
2020
2030
2040
2050
IMAGE
PROMETHEUS
TIAM-ECN
LIMES
OSeMBE
Euro-Calliope
PRIMES
REMIND
WITCH
MEESA
Historical
median

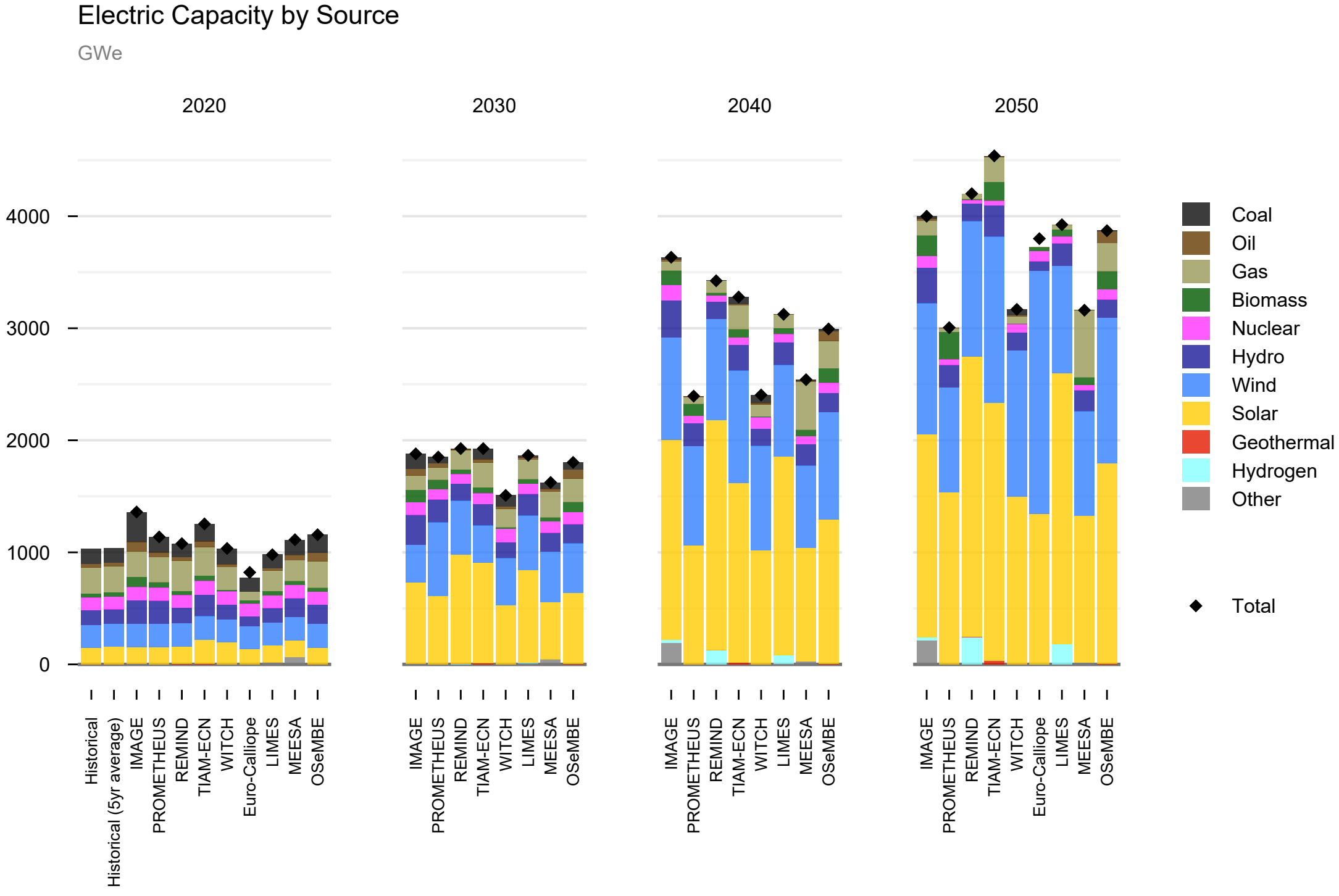

Share of electricity and hydrogen in final energy consumption
%
70
60
50
40
30
20
10
0
Historical
2000
2010
2020
2030
2040
2050
IMAGE
PROMETHEUS
TIAM-ECN
median
Electricity
PRIMES
REMIND
WITCH
Historical
Hydrogen

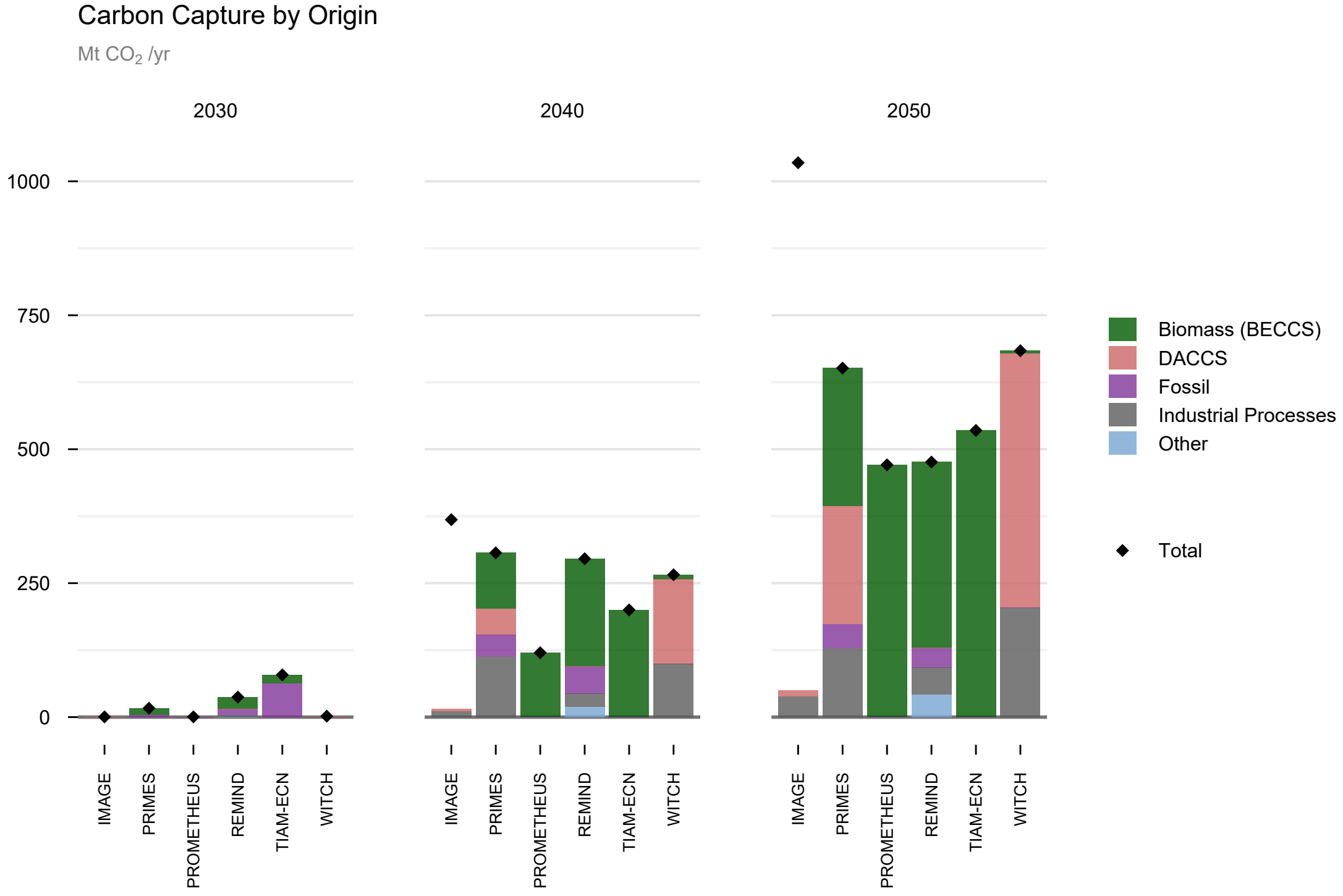

(a) Carbon-based final energy
EJ/yr
2020
2030
2040
2050
40
30
20
10
0
IMAGE
PRIMES
REMIND
TIAM-ECN
WITCH
PROMETHEUS
(b) Net energy imports
EJ/yr
2020
2030
2040
2050
40
30
20
10
0
Historical
IMAGE
PRIMES
PROMETHEUS
REMIND
TIAM-ECN
WITCH
Synfuel - Gases
Synfuel - Liquids
Biomass - Gases
Biomass - Liquids
Biomass - Solids
Fossils - Gases
Fossils - Liquids
Fossils - Solids
Electricity
Hydrogen
Liquids (Synfuel)
Liquids (bio-based)
Biomass
Gas
Oil
Coal


(a) Emissions per sector
Mt CO_2e /yr
Total net Kyoto gases
2020
2030
2040
2050
4000
3000
2000
1000
0
-1000
Historical
Historical (5yr average)
IMAGE
PRIMES
PROMETHEUS
REMIND
TIAM-ECN
WITCH
Demand - Bunkers
non-CO2
other CO2
Demand - AFOFI
Demand - Buildings
Demand - Industry
Industrial Processes
Demand - Transport
Energy Supply - Other
Energy Supply - Solids
Energy Supply - Liquids
Energy Supply - Gases
Energy Supply - Heat
Energy Supply - Electricity
Geological Carbon Removal
LULUCF
(b) Carbon Capture
Mt CO_2 /yr
1000
800
600
400
200
0
Historical
2010
2020
2030
2040
2050
IMAGE
PRIMES
PROMETHEUS
REMIND
TIAM-ECN
WITCH
median


(a) Carbon Price
€/tCO2
800
600
400
200
0
Historical
2020
2030
2040
2050
IMAGE
PROMETHEUS
TIAM-ECN
LIMES
OSeMBE
PRIMES
REMIND
WITCH
MEESA
median
(b) Mitigation costs
% of GDP
1.0
0.5
0.0
WITCH
REMIND
PRIMES
TIAM-ECN
PROMETHEUS
IMAGE
n = 6
Maximum
75th percentile
Median
25th percentile
Minimum


Electricity Generation by Source
TWh/yr
2020
2030
2040
2050
9000
6000
3000
0
Historical
Historical (5yr average)
IMAGE
PRIMES
PROMETHEUS
REMIND
TIAM-ECN
WITCH
Euro-Calliope
LIMES
MEESA
OSeMBE
Coal
Oil
Gas
Biomass
Nuclear
Hydro
Wind
Solar
Geothermal
Hydrogen
Other
Total


Electric Capacity by Source
GWe
2020
2030
2040
2050
4000
3000
2000
1000
0
Historical
Historical (5yr average)
IMAGE
PROMETHEUS
REMIND
TIAM-ECN
WITCH
Euro-Calliope
LIMES
MEESA
OSeMBE
Coal
Oil
Gas
Biomass
Nuclear
Hydro
Wind
Solar
Geothermal
Hydrogen
Other
Total


Carbon Capture by Origin
Mt CO_2 /yr
2030
2040
2050
1000
750
500
250
0
IMAGE
PRIMES
PROMETHEUS
REMIND
TIAM-ECN
WITCH
Biomass (BECCS)
DACCS
Fossil
Industrial Processes
Other
Total